Team APCS ⊕ CNTN

AIC26-Retrieval: Evidence-First Video Retrieval with Guarded AI Agents

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INPUT

Corpus: 1,478 broadcasts, 298 traffic cameras and 12 cycling videos



2.21M InfoShot++ keyframes · 496K clips

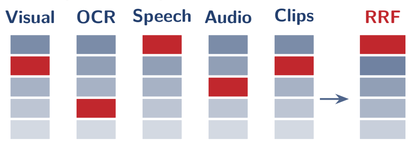
Queries: text or a clip

- **T-KIS:** a described moment, plus hints
- **V-KIS:** a short preview clip
- **QA:** the moment *and* an answer
- **TRAKE:** $E_1 \to E_2 \to E_3 \to E_4$, in order

Queries are in Vietnamese and every answer goes to the organisers' DRES server.

A Multi-Channel Search

Routing to visual (VI→EN), OCR, speech, audio and clip channels; topic folders pushed into the indexes.



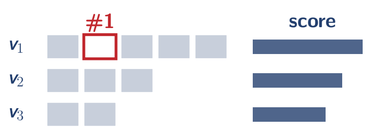
$$R(f) = \sum_{c:\, f \in L_c} \frac{w_c}{60 + r_c(f)}$$

$r_c(f)$: rank of frame f in channel c

Ranks are fused; cosine scores from different embedding spaces are *never* added.

B Groups & TRAKE

Frames are grouped by video and shown in time order.



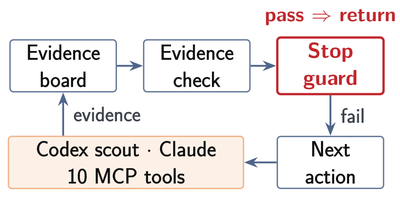
Video score: best frame, top-3 mean and temporal support; distant top frames flag *ambiguity*.



TRAKE: shared context per event and temporal NMS, then an **exact DP** picks the best increasing-time chain; a second pass fills missing events.

C Guarded Agents

AGENT mode runs beside every search on a query-local evidence board.



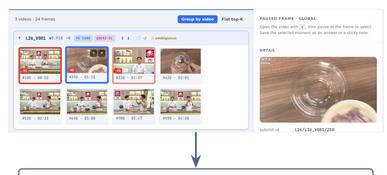
Each query clause is checked against the evidence as *supported*, *refuted* or *unknown*.

Agents search only while the check fails; the console never waits, and a new search cancels the run.

Codex: GPT-6.1 Sol · Claude: Opus 5.5

OUTPUT

Operator console: grouped videos, evidence and streamed agent picks



Submit guard → DRES

58.1% R@1
strict, vs 15.1% for retrieval only

86 AIC 2026 TEST queries, ±1 s

candidates seed the evidence board · async, never blocks the console

1. Introduction

AIC26-Retrieval is the interactive video retrieval system our team built for AI Challenge HCMC 2026. It indexes 2.21M adaptive keyframes and 496K multi-scale clips, routes each Vietnamese query to visual, OCR, speech, audio and clip channels, fuses them by rank and groups results by video. Task modules cover T-KIS hints, V-KIS sketches, QA answers and ordered TRAKE chains; AI agents search in parallel, and a submit guard checks every answer before it reaches DRES.

The challenge

- Queries mix appearance, actions, on-screen text, speech and sounds; no single model covers all of them.
- Bulletins repeat near-identical scenes, so the right video is not yet the right moment.
- The answer is an exact frame, chosen and submitted by one person against the clock.

▶ *Design principle: UI > Method > Model – minimise wrong submissions and navigation time.*

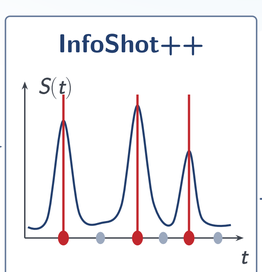
2. Methodology

(a) Offline indexing (before the contest)



Raw videos
1,478 broadcasts
298 traffic cams
12 cycling races

InfoShot++



content-change events, native-rate refinement, gap filling

Frame registry

video	frame	t (s)
L27_V014	634	25.4
L27_V014	675	27.0
…	…	…

2,212,383 keyframes on one time axis

PE-Core · Qwen3-VL → Milvus vectors
TARA clips → Milvus vectors
HunyuanOCR → Elasticsearch
PhoWhisper ASR → Elasticsearch
CED · GLAP audio → Elasticsearch

(b) Online search, verification and submission

kNN and text search

Query
"Trong một lễ hội có nhiều người hóa trang…"
+ hints, a sketch or a clip

Query parser
task type
channels, weights
VI→EN, scope

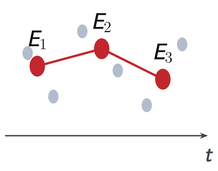
Vis. Clip OCR ASR Aud. → RRF → **Video groups**

seed

T-KIS / V-KIS
exact paused frame; sketch as image query

QA copilot
answer + evidence frame, human check

TRAKE exact DP



max chain with $t_1 < t_2 < t_3$

AI agents
evidence board → clause check
supported / refuted / unknown
Codex scout · Claude investigator
10 MCP tools, ≤ 240 s

Operator console
groups · timeline · player

Submit guard
dedup · format → **DRES**

Figure 1. Methodology. (a) Offline, InfoShot++ selects keyframes where content changes and a registry puts every index on one time axis. (b) Online, a query is parsed, searched over five channels, fused by rank and grouped by video; task modules, AI agents and the operator turn candidates into a guarded DRES submission.

Indexing.
InfoShot++ tracks global, tile and on-screen-text changes, motion and a SigLIP probe, refines frames around events at the native frame rate and fills temporal gaps; the organisers' 382,299 keyframes remain a second profile. Vectors live in Milvus; OCR, speech and audio tags in Elasticsearch.

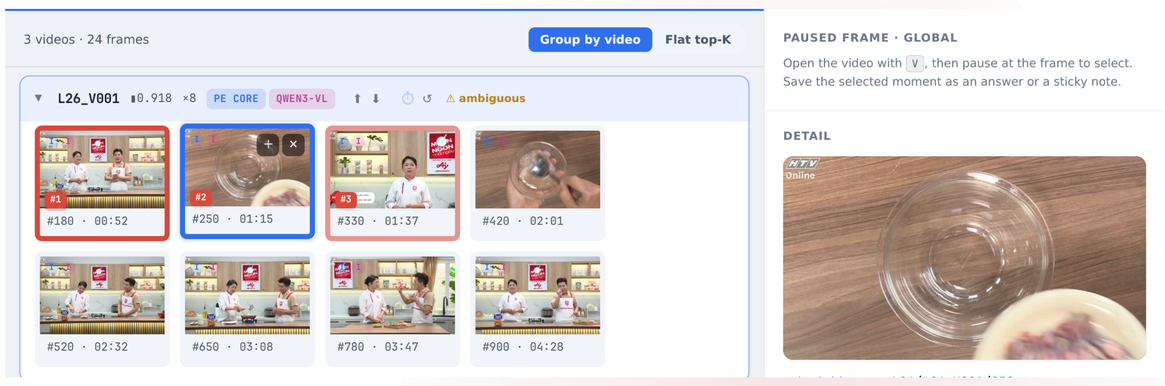
Search and ranking.
A heuristic router or a DeepSeek V4.1 Flash parser sets channels and weights; visual text is translated to English while OCR and speech keep Vietnamese, and topic folders or camera banners narrow the scope. Weighted RRF fuses ranks only, and each video is scored by its best frame, top-three mean and temporal support.

Task modules.
T-KIS appends hints, and Progressive Hint Memory keeps a ledger of released hints with video-level evidence across turns; the operator submits the exact paused frame; V-KIS searches a 16:9 sketch as an image; the QA copilot proposes answers with evidence frames; TRAKE composes shared context into each event and solves an exact DP over increasing-time chains, with a second in-video pass. In the preliminary round, a 100-answer generator spends the rank budget.

Guarded agents.
Retrieval candidates seed a query-local evidence board, and each query clause is checked as *supported*, *refuted* or *unknown*. Only while this check fails do Codex scout and Claude investigator search, inspect frames and compare candidates; the console never waits for them, and a new search cancels the run.

2.21M	496K	5	1.6 s	10
InfoShot++ keyframes	TARA clips	search channels	median retrieval latency	MCP tools for agents

3. Operator Console and AI Agents

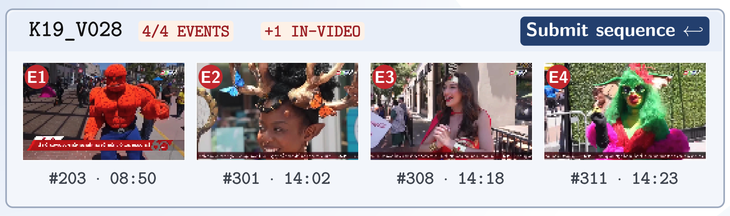


Results grouped by video with the top three frames marked (left); frame detail, matched channels and submission ID (right).

- **One keyboard-first screen**: grouped and flat views, a multi-track timeline (keyframes, speech, OCR, audio) and a player that captures the exact paused frame.
- **Submit guard**: shows the exact DRES payload and the open task, and blocks duplicates, QA answers without text, and TRAKE sequences that mix videos or break time order.
- **AI agents**: Codex scout and Claude investigator search through ten MCP tools while the evidence check fails; their picks stream into an Agents strip, with no shell and no credentials.

4. Case Study

TRAKE, trake-01: costume appearances, in order.
"Trong một lễ hội có nhiều người hóa trang, lần lượt đưa ra khoảnh khắc đầu tiên những người hóa trang các nhân vật sau xuất hiện…" The top-ranked video covers all four events, one filled by the in-video pass, and the operator submits the whole sequence with one key.

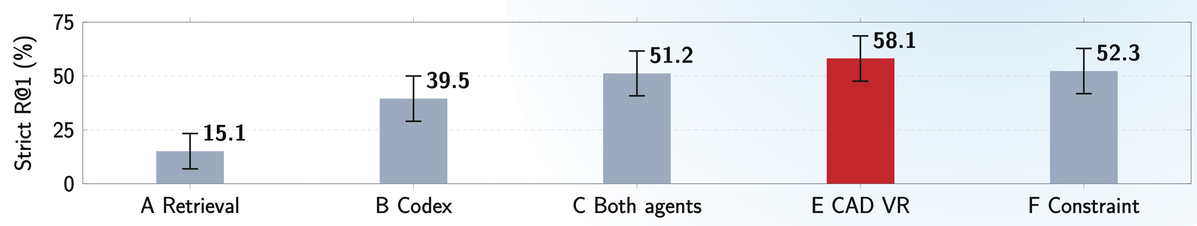


Result card for K19_V028: badges *4/4 events* and *+1 in-video*; E1 to E4 at 08:50, 14:02, 14:18 and 14:23, in time order.

T-KIS, tkis-query-23: an action, not an object.
A worker towing a bin by motorcycle drops a white bag. Frame search (PE) missed the action, while clip retrieval (TARA) returned the right video and segment first from hint 1 (N061_V002, 289.6 s); later hints confirmed it.

5. Results

86 AIC 2026 TEST queries (59 T-KIS, 19 QA, 8 TRAKE), fresh live runs. *Strict* R@1: exact moment within ±1 s, plus the answer (QA) or full chain (TRAKE).



Configuration	R@1	R@5	MRR	p50 (s)	Agents
A Retrieval only	15.1	33.7	.233	1.6	0
B Always Codex	39.5	68.6	.529	50.5	1.00
C Always both	51.2	**72.1**	.612	51.2	2.00
E CAD VR (ours)	**58.1**	67.4	**.630**	72.6	1.95
F CAD VR, constraint-level	52.3	64.0	.572	84.2	1.99

Grouped by video (R@1, %)

	Strict	Group	Video
A Retrieval	15.1	43.0	45.3
E CAD VR	**58.1**	**81.4**	**86.0**

With only the first hint (24 queries), video R@1 is 83.3% for CAD VR (constraint-level) vs 41.7% for retrieval.

Progressive hints (60 held-out queries, PE+Qwen)

Method	MRR	p50 (s)
Independent hints (RRF)	.666	1.11
Cumulative query	.691	1.13
Progressive Hint Memory	.690	3.88

PHM matches cumulative search while keeping per-hint evidence.

▶ *Repeated ablation: CAD VR 58.1% vs 50.0% for always-both with final verification; not significant ($p = 0.092$), and slower.*

6. Lessons and Conclusion

- **Search by hand beside the agents**, in a second tab with agents off, so their run is not cancelled.
- **Use clips for action queries**, and **check evidence, not scores**: OCR, speech and neighbouring frames separate near-identical scenes.

▶ *Rank fusion, video grouping and a guarded submission path let one operator go from a partial description to a verified moment. Next: measure operator time-to-correct and wrong-submission rate.*

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